Task 2: Multiple sequence alignment:

I took the sequence of protein showing high percentage identity with the my query sequence in FASTA formats. Perform the multiple sequence alignment using clustal omega. The sequence is from *Homo sapiens, Ailuropoda melanoleuca, Oryctolagus cuniculus, Mus musculus*, and *Pan troglodytes*.

```
AAA35952.1          ------------------------------------------------------------     0
EFB18430.1          MVHFTAEEKAAVVSLWARVNVELVGGEVLGRLLVVYPWTQRFFDSFGNLSSESAIMGNPK     60
NP_001129304.1      ------------------------------------------------------------     0
NP_001300972.1      ------------------------------------------------------------     0
NP_058652.1         ------------------------------------------------------------     0


AAA35952.1          ---------------------------------------------------MVHLTPEEK     9
EFB18430.1          VKAHGKKVLTSFGNAVKHMDDLKDTFAELSELHCDKLHVDPENFKQPQRDTMVHLTGEEK     120
NP_001129304.1      ---------------------------------------------------MVHFTAEEK     9
NP_001300972.1      ---------------------------------------------------MVHLSSEEK     9
NP_058652.1         ---------------------------------------------------MVHLTDAEK     9
                                                                       ***::  **

AAA35952.1          SAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKVKAHGKKVL     69
EFB18430.1          AAVTGLWSKVNVDEVGGEALGRLLVVYPWTQRFFDSFGDLSTPDAVMNNPKVKAHGKKVL     180
NP_001129304.1      AAVTSLWSKMNVEEAGGEALGRLLVVYPWTQRFFDSFGNLSSPSAILGNPKVKAHGKKVL     69
NP_001300972.1      SAVTALWGKVNVEEVGGEALGRLLVVYPWTQRFFESFGDLSSAHAVMSNPKVKAHGKKVL     69
NP_058652.1         SAVSCLWAKVNPDEVGGEALGRLLVVYPWTQRYFDSFGDLSSASAIMGNPKVKAHGKKVI     69
                    :**: **.*:* :*.*****************:*:***:**:  *::.***********:

AAA35952.1          GAFSDGLAHLDNLKGTFATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQP     129
EFB18430.1          NSFSEGLKNLDNLKGTFAKLSELHCDKLHVDPENFKLLGNVLVCVLAHHFGKEFTPQVQA     240
NP_001129304.1      TSFGDAIKNMDNLKPAFAKLSELHCDKLHVDPENFKLLGNVMVIILATHFGKEFTPEVQA     129
NP_001300972.1      AAFSEGLNHLDNLKGTFAKLSELHCDKLHVDPENFRLLGNVLVVVLSHHFGKEFTPQVQA     129
NP_058652.1         TAFNEGLKNLDNLKGTFASLSELHCDKLHVDPENFRLLGNAIVIVLGHHLGKDFTPAAQA     129
                     :*.:.: ::**** :**.****************:****.:* :*. *:**:*** .*

AAA35952.1          HLSVVWLMPWPTT-PGQRAGLCAGP-SLWQIIHPTSAGCLSESGGWCG    175
EFB18430.1          AYQKVVAGVANALAHKYH------------------------------    258
NP_001129304.1      AWQKLVSAVAIALAHKYH------------------------------    147
NP_001300972.1      AYQKVVAGVANALAHKYHXDLFPSAKNYGDIMKPLEHL---TSG----    170
NP_058652.1         AFQKVVAGVATALAHKYH------------------------------    147
                      . :      :     :
```

The multiple sequence alignment (MSA) results in the source show a comparison between three protein sequences: AAA35952.1, EFB18430.1, and NP_001300972.1, NP_001300972.1, and NP_058652.1. This alignment identifies conserved evolutionary patterns and structural similarities across these protein variants, which are characteristic of the hemoglobin beta (HBB) family discussed in our previous conversation.

Interpretation of Alignment Symbols

The consensus line at the bottom of each alignment block uses standard bioinformatics symbols to denote the level of conservation:

Asterisk (*): Indicates positions where all three sequences have an identical amino acid, representing high evolutionary conservation.

Colon (:): Indicates positions where the amino acids have strongly similar biochemical properties.

Period (.): Indicates positions where the amino acids have weakly similar properties.

Blank Space: Represents a position with no conservation, where the sequences differ significantly.

Explanation of results:

The alignment confirms that these three sequences share a common ancestral domain related to the hemoglobin beta chain but differ in their terminal extensions. The high degree of central conservation indicates that the core structural and functional properties, specifically the globin fold, are maintained across these different variants, despite the variations observed at their start and end points.